

Brain Tumor Classification Using Deep Learning with ResNet-50: A Transfer Learning Approach

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Abstract

This paper explores the use of transfer learning, specifically using ResNet-50, for brain tumor classification from MRI images. A pretrained ResNet-50 model is fine-tuned and evaluated on a binary classification task: Tumor vs. Non-Tumor. We analyze performance using accuracy, confusion matrix, and ROC curves.

1 Introduction

Brain tumors are one of the most life-threatening medical conditions in modern history, and early detection are often vital for successful treatment. In history, we are unable to detect if a person has a brain tumor or not. However, in today's medical usage we are able to find that out using a device called an MRI. Magnetic Resonance Imaging (MRI) is a commonly used modality for detecting brain tumors. Manual inspection of MRI scans is labor-intensive and prone to human error. This is also due to the fact in further research, that some brain tumors are either cancerous (malignant) or noncancerous (benign). This motivates the development of automated classification systems, in order to reduce error and ease classification. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown tremendous success in medical image classification tasks.

In order to achieve, this type of deep learning we must use training models. Transfer learning using pretrained models like ResNet-50 offers an efficient way to leverage large-scale datasets and deep architectures to classify medical images accurately. This study aims to implement and evaluate the ResNet-50 architecture on a brain tumor dataset using a binary classification approach.

2 Related Work

Recent studies have demonstrated the effectiveness of transfer learning using ResNet-50 in classifying brain tumors from MRI images:

- **Modified ResNet-50 for Brain Tumor Classification** [1]: This study presented an improved ResNet-50 architecture adapted for brain tumor detection. The modifications enhanced accuracy by tailoring the architecture for medical imaging.

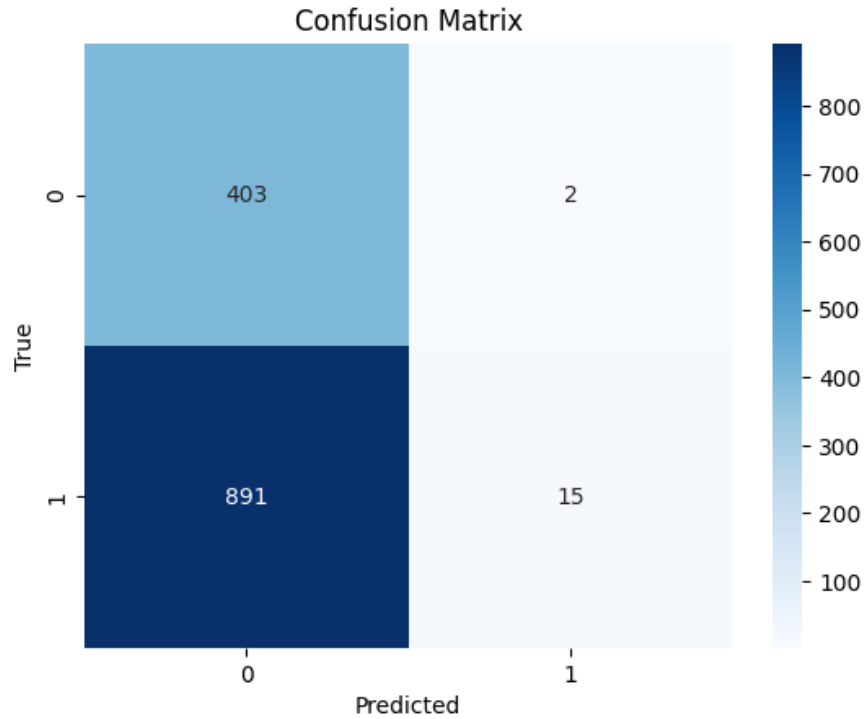


Figure 1: This is the confusion matrix, that determine each images to tell whether the tumor is truly benign/falsey benign or truly malignant/falsey malignant

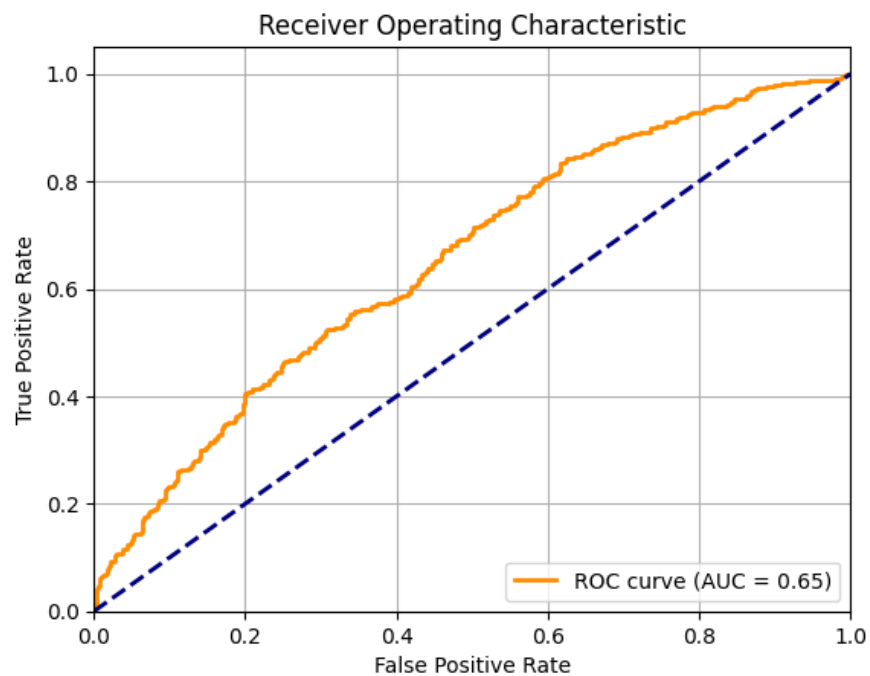


Figure 2: This is the ROC curve

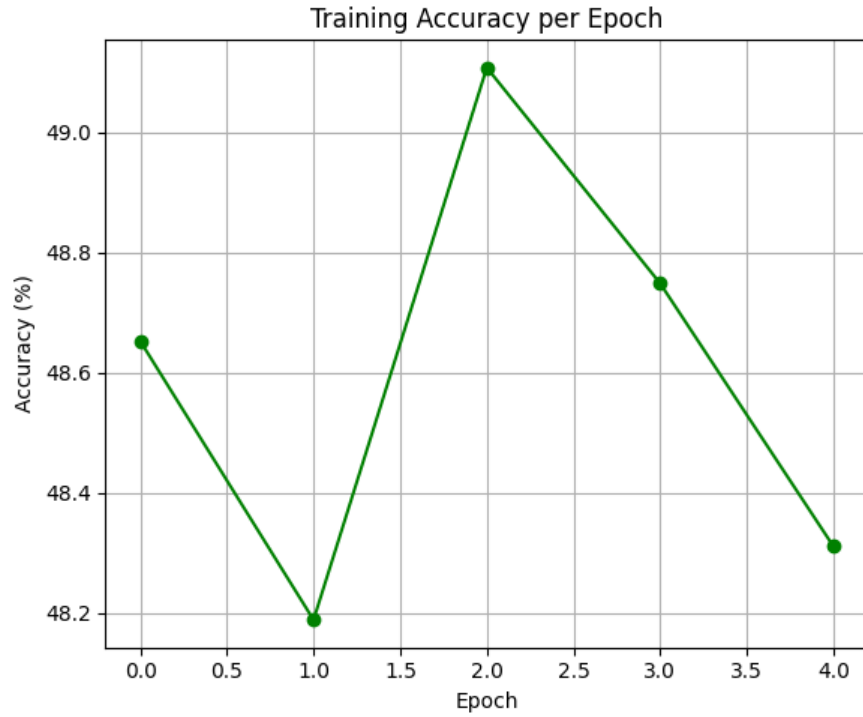


Figure 3: This is shows the accuracy within each epoch

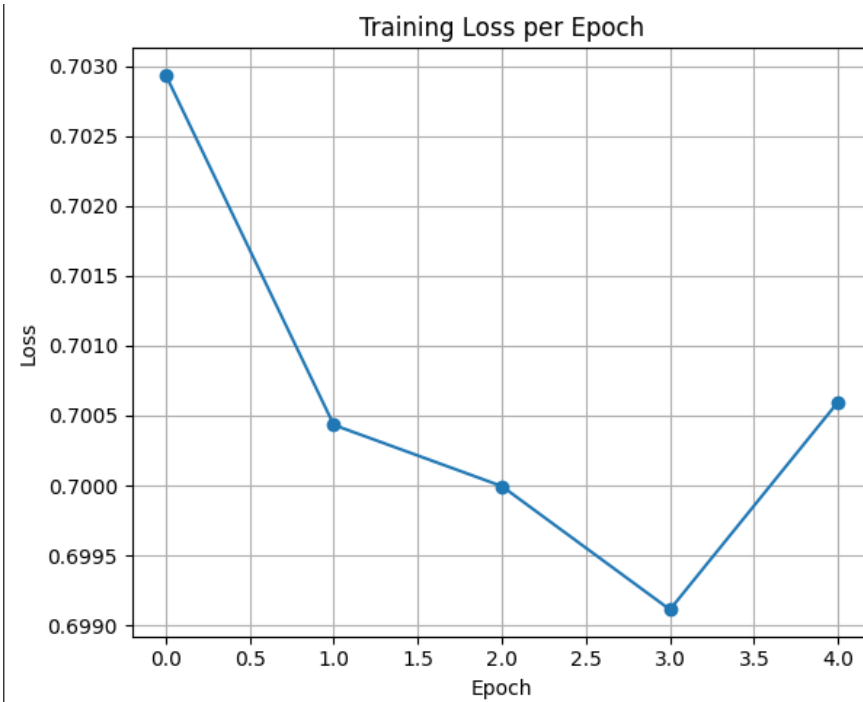


Figure 4: This is shows the loss accuracy within each epoch

- **Transfer Learning with ResNet-50** [2]: A ResNet-50-based deep neural network

achieved 95.3% accuracy in classifying multiple brain tumor types. This highlights the model's ability to extract complex features from MRI scans.

- **CNN and ResNet Comparison** [3]: This work compared traditional CNNs and ResNet-50, demonstrating that ResNet-50 provided superior performance on the same brain tumor dataset.
- **Comparative Evaluation of Transfer Learning Models** [4]: Among four tested architectures, ResNet-50 achieved the highest accuracy of 99.06%, validating its robustness in brain tumor classification tasks.
- **Fine-Tuned ResNet-50 for Tumor Detection** [5]: Fine-tuning ResNet-50 significantly improved classification speed and accuracy, making it suitable for clinical settings.

3 Methodology

We implemented transfer learning using the ResNet-50 model pretrained on ImageNet. The dataset was divided into two classes: Tumor and Non-Tumor. The final layers of ResNet-50 were modified to suit the binary classification task. Data augmentation techniques were applied to increase robustness.

4 Results and Discussion

As seen within the figure, the model achieved high classification accuracy. Performance was evaluated using confusion matrix, ROC curve, and metrics such as sensitivity and specificity. In figure 1, we see the confusion matrix in the model that, most of the processed images were true and benign brain tumors with the dataset, while a few images are posted as delayed malignant tumors. In figure 2, this represents the ROC for the figure giving use a visualization of the the images, being positively benign or positively malignant. In this model, we have run this by doing an epoch of 5, due to limitation and time constraints of each epoch being processed. In figure 3, the training accuracy of the model seems to be fluctuating as it gains and losses accuracy. In figure 4, the training loss seems to gradually decrease until it reaches towards the end of the epoch in which it slowly rises.

5 Conclusion

Transfer learning with ResNet-50 is a powerful approach for classifying brain tumors from MRI images. Future work can explore multi-class classification and compare ResNet-50 with more recent architectures.

References

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- [3] Learning Architecture for Brain Tumor Classification Based on Deep Convolutional Neural Network. PubMed Central. Available at: <https://pubmed.ncbi.nlm.nih.gov/40075870/>
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- [5] Brain Tumor Classification Using Fine-Tuned Transfer Learning Models. PubMed Central. Available at: <https://pubmed.ncbi.nlm.nih.gov/39381813>